

Dimension	Bicritical	Tricritical	Tetracritical	Pentacritical	Hexacritical
3	0.0587				
2.8	0.0900				
2.6	0.1279	0.0084			
2.4	0.1715	0.0238	0.0042		
2.2	0.2190	0.0500	0.0173	0.0070	
2.1	0.2430	0.0651	0.0290	0.0151	0.0088
2.05	0.2555	0.0730	0.0332	0.0201	0.0129
2.01	0.2646	0.0812	0.0387	0.0209	0.0150
2.001	0.2677	0.0821	0.0639	0.0388	0.0211

Table 1: The Anomalous Dimension for different effective dimensions ranging from $D_{eff} = 3$ to $D_{eff} = 2.001$, for the first 5 multi-critical points. Notice that the Sak points occur when $D_{eff} \rightarrow 2$, that is, corresponding to the last row.

dimension	3	2.8	2.6	2.4	2.2	2.1	2.05	2.01	2.001
$y_{2,1}$	1.547	1.430	1.304	1.160	1.001	0.929	0.869	0.791	0.815
$y_{3,1}$			1.985	1.958	1.919	1.898	1.904	1.889	1.892
$y_{3,2}$			0.932	0.839	0.701	0.599	0.500	0.440	0.413
$y_{4,1}$				1.995	1.973	1.958	1.952	1.946	1.913
$y_{4,2}$				1.460	1.361	1.193	1.022	0.824	0.983
$y_{4,3}$				0.662	0.569	0.482	0.392	0.290	0.446
$y_{5,1}$					1.992	1.976	1.973	1.967	1.949
$y_{5,2}$					1.724	1.763	1.811	1.868	1.835
$y_{5,3}$					1.136	1.031	0.890	0.725	0.731
$y_{5,4}$					0.479	0.416	0.344	0.386	0.260
$y_{6,1}$						1.992	1.982	1.979	-
$y_{6,2}$						1.832	1.859	1.910	-
$y_{6,3}$						1.388	1.241	0.914	-
$y_{6,4}$						0.905	0.797	0.551	-
$y_{6,5}$						0.365	0.311	0.179	-

Table 2: The RG eigenvalues for the first five multicritical models, for the respective effective dimensions discussed. In the row corresponding to $D_{eff} = 2.001$, we could not find the spike corresponding to the hexacritical point, therefore the corresponding eigenvalues are missing.

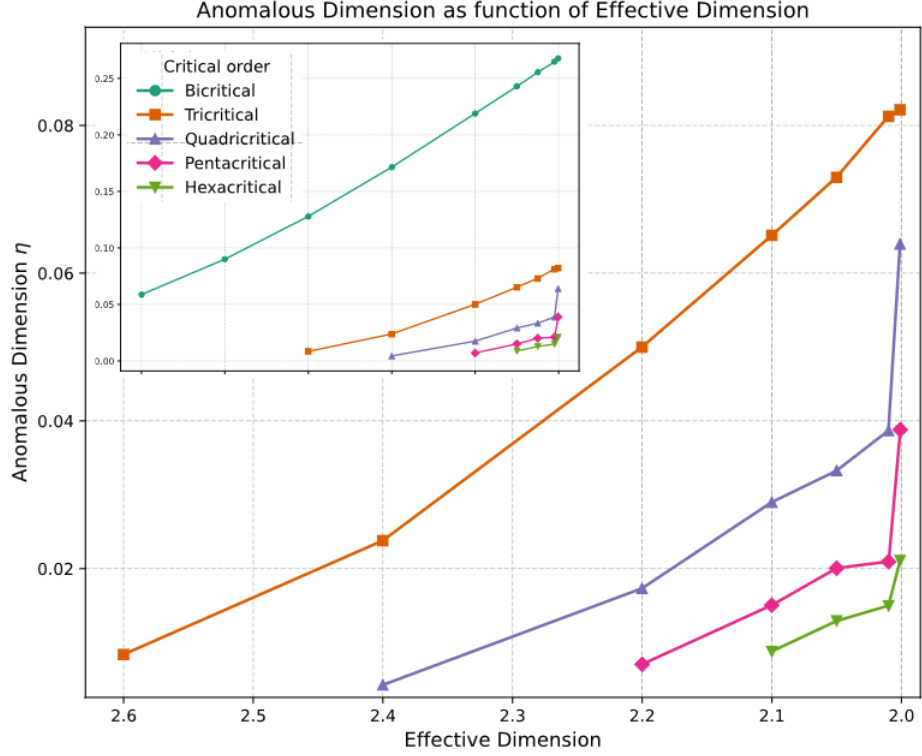


Figure 0.1: The anomalous dimensions for the first five multicritical models, as function of the anomalous dimension. In the inset, the zoomed-out version of the graph, displaying also the result for the bicritical model, whose anomalous dimension lays significantly above the other models.

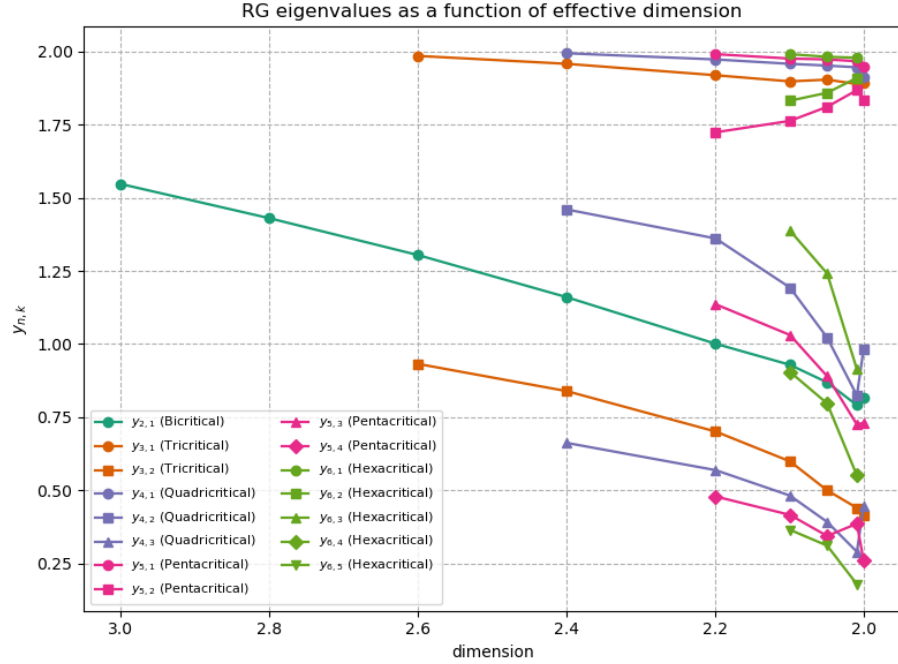


Figure 0.2: The RG eigenvalues as function of the effective dimension, for the first five multicritical points.

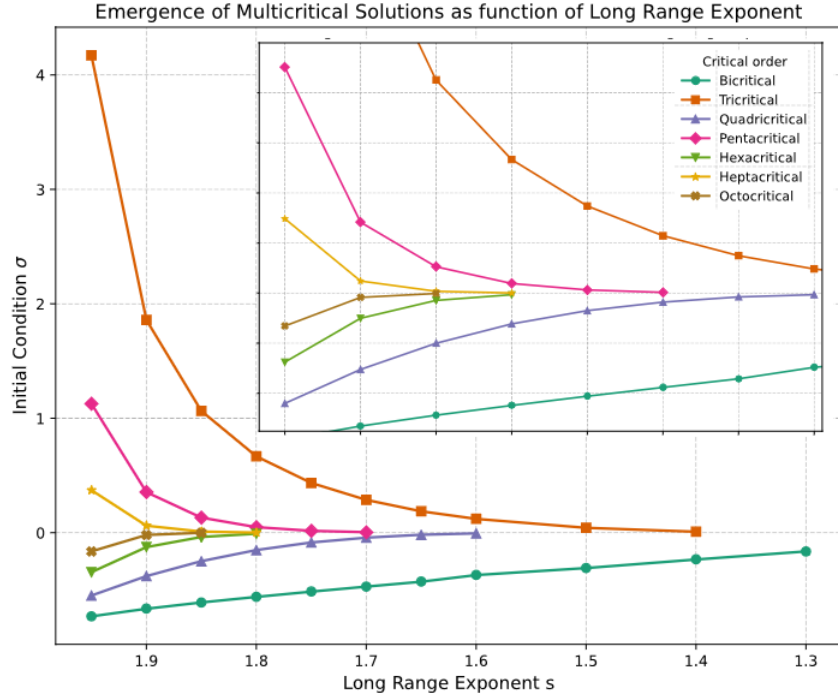


Figure 0.3: The emergence of multicritical points as a function of the long range exponent s . The y -axis displays the initial condition $\tilde{U}_t^{(1)}(\bar{\rho} = 0) = \sigma$ that triggers the RG flow towards the particular multicritical point, whereas the x -axis displays the particular long-range exponent at which this solution σ exists.

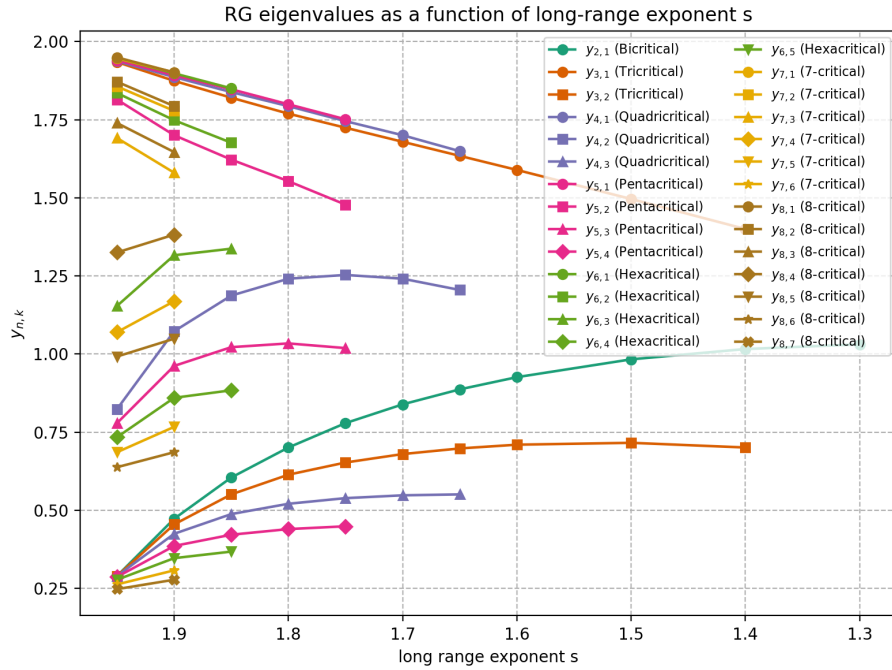


Figure 0.4: The RG eigenvalues for the first eight CFT multicritical models as function of the long-range exponent s .

s factor	1.3	1.4	1.5	1.6	1.65	1.7	1.75	1.8	1.85	1.9	1.95
$y_{2,1}$	1.031	1.016	0.983	0.926	0.887	0.839	0.779	0.701	0.605	0.473	0.290
$y_{3,1}$		1.400	1.496	1.589	1.634	1.679	1.724	1.769	1.820	1.874	1.934
$y_{3,2}$		0.701	0.716	0.710	0.698	0.680	0.653	0.614	0.551	0.455	0.290
$y_{4,1}$					1.649	1.700	1.745	1.793	1.838	1.886	1.937
$y_{4,2}$					1.205	1.241	1.253	1.241	1.187	1.073	0.824
$y_{4,3}$					0.551	0.548	0.539	0.521	0.488	0.425	0.290
$y_{5,1}$							1.751	1.799	1.847	1.892	1.940
$y_{5,2}$							1.478	1.553	1.622	1.700	1.814
$y_{5,3}$							1.019	1.034	1.022	0.962	0.779
$y_{5,4}$							0.449	0.440	0.422	0.386	0.287
$y_{6,1}$									1.850	1.898	1.946
$y_{6,2}$									1.676	1.748	1.835
$y_{6,3}$									1.337	1.316	1.154
$y_{6,4}$									0.884	0.860	0.734
$y_{6,5}$									0.368	0.347	0.278
$y_{7,1}$										1.901	1.946
$y_{7,2}$										1.778	1.856
$y_{7,3}$										1.580	1.691
$y_{7,4}$										1.169	1.070
$y_{7,5}$										0.767	0.686
$y_{7,6}$										0.308	0.263
$y_{8,1}$										1.901	1.949
$y_{8,2}$										1.793	1.871
$y_{8,3}$										1.646	1.739
$y_{8,4}$										1.382	1.325
$y_{8,5}$										1.049	0.992
$y_{8,6}$										0.686	0.638
$y_{8,7}$										0.278	0.248

Table 3: The RG eigenvalues for the first eight CFT multicritical models, as function of the long-range exponent s for a long-ranged model.